GYANMANJARI INNOVATIVE UNIVERSITY (GMIU)

Project Report: Rule-Based Central Library Inquiry Chatbot

Department of Information Technology • Academic Artificial Intelligence Assignment

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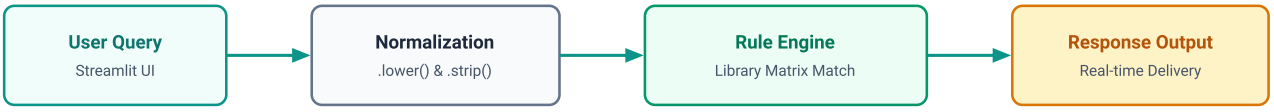
Application: [Your Live Streamlit Link / Web URL]

Tech Stack: Python 3.x, Streamlit Cloud, Git / GitHub

1. PROJECT ABSTRACT & OBJECTIVES

The objective of this project is to implement an automated conversational assistant tailored specifically for university library services. Built with Python using deterministic `if-elif-else` conditional logic, the chatbot addresses routine student inquiries regarding borrowing limits, operating hours, overdue return fines, catalog locations, and digital e-journal access. This system relieves circulation desk personnel from repetitive administrative tasks while providing instantaneous answers.

2. SYSTEM FLOW & ARCHITECTURE



3. ACADEMIC STUDENT CREDENTIALS

Sr. No.	Student Full Name	Enrollment Number	Project Role
01	Sarvaiya Aryarajsinh	241105060118	Rule base definition, string preprocessing & test logs
02	Gohil Harshrajsinh	241105060153	Streamlit frontend interface, deployment & report design

4. VERIFICATION & SAMPLE CONVERSATION TEST LOG

The table below records actual test inquiries evaluated against the Central Library rule-based engine:

#	Student Input Query	Chatbot Answer Output	Matched Category
Q1	"Hello / Good morning"	Hello! Welcome to the Central Library Helpdesk. How can I assist you with book searches, borrowing rules, or fines?	Greeting
Q2	"What are the library timings?"	Library Operating Hours: Monday to Saturday 8:00 AM - 8:00 PM; Reading Hall open until 10:00 PM daily.	Working Hours
Q3	"How many books can I borrow?"	Diploma and UG students may borrow up to 3 books for 14 calendar days using a valid student ID card.	Borrowing Rules
Q4	"What is the fine for late return?"	Overdue Fine: ₹2 per book per day for the first 7 days, and ₹5 per day thereafter. All dues must be cleared.	Fine Policy
Q5	"Can I renew my books?"	Books can be renewed up to 2 consecutive times provided no other student has placed a reservation hold.	Renewal Desk
Q6	"Do you have IEEE journals or Wi-Fi?"	We offer access to DELNET, IEEE Xplore, and NDLI across 40 PC terminals on the 2nd Floor Digital Library.	Digital Access
Q7	"Where can we do group study?"	Quiet reading halls have 300 seats, and 3 collaborative discussion rooms are available on the 3rd floor.	Study Rooms
Q8	"Can I order food here?"	I'm sorry, I couldn't locate that topic. Ask me about borrowing limits, timings, late fines, or e-journals.	Fallback Handler

5. CORE RULE IMPLEMENTATION SNIPPET (`LIBRARY_APP.PY`)

```
      def get_library_response(user_input: str) -> str: text = user_input.lower().strip() if text
in ["hi", "hello", "good morning"]: return "Welcome to the Central Library Helpdesk. How can I
assist you?" elif "timing" in text or "hour" in text or "open" in text: return "Library is open
Mon-Sat 8:00 AM - 8:00 PM; Reading Hall until 10:00 PM." elif "borrow" in text or "issue" in text
or "limit" in text: return "UG/Diploma students can borrow up to 3 books for 14 calendar days."
elif "fine" in text or "late" in text: return "Late fine: ₹2/day for first 7 days, ₹5/day
thereafter." else: return "I couldn't match that. Ask about timings, book issues, fines, or
journals."
```

6. CONCLUSION

The Central Library Inquiry Chatbot satisfies all curriculum criteria for rule-based systems using structured Python `if-elif-else` evaluation. It delivers immediate, deterministic answers for recurring library inquiries without external API costs or server complexity.